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CONFERENCE PROCEEDINGS

Correction of Patient–Ventilator Discordance

John Davies,* Elias Pratt, and Craig R. Rackley

Abstract

Optimizing the interaction between the patient and the ventilator should be a cornerstone for clinicians to pursue. Disruptions in these interactions can lead to what is called patient–ventilator discordance. Patient–ventilator discordance is a mismatch between what the patient desires in a breath and what the ventilator provides. Asynchrony is also a common term used to describe these unhealthy interactions. Different types of asynchronies make up the more global term of patient–ventilator discordance. Asynchronies can occur at the onset of a breath (trigger asynchrony), during breath delivery (flow asynchrony), or at the end of a breath (cycle asynchrony). Prompt identification and correction of these asynchronies is pivotal to prevent overuse of sedation, extended duration of mechanical ventilation, and perhaps an increased mortality risk. Clinicians must also take into consideration the phenomenon of ventilator-related dyspnea. Dyspnea is defined as a subjective feeling of breathing discomfort. In a ventilated patient dyspnea often goes unrecognized, as there

Mr. Davies, Dr. Pratt, and Dr. Rackley are affiliated with Duke University Medical Center, Durham, North Carolina, USA.

*Address correspondence to: *John Davies, MA, RRT, Duke University Medical Center, Box 3911, Duke Hospital, Durham, NC 27710, USA,* Email: john.davies@duke.edu

is a communication barrier. Similar to the sensation of pain, recognizing and managing dyspnea is vital in mechanically ventilated patients.

Keywords: Patient–ventilator discordance, trigger asynchrony, flow asynchrony, work shifting, cycle asynchrony, reverse triggering, dyspnea

Introduction

Mechanical ventilation is a mainstay in the ICU for patients with various conditions. Appropriate communication between the patient and ventilator is essential to optimize patient comfort and reduce the overuse of sedation. When this communication is suboptimal, patient–ventilator discordance (PVD) can occur. PVD is a mismatch between the patient's neural inspiratory efforts and what the mechanical ventilator provides in terms of timing and breath delivery. Often it is described as the "patient fighting with" or "bucking" the ventilator. Discordance can result from either under- or over-assistance from the ventilator and manifests in various different asynchronies. It is a common phenomenon estimated to occur in up to 80% of patients at some point during their ICU stay.[1] Thille et al reported that ∼25% of mechanically ventilated subjects experienced a high frequency of asynchronous events.[2] Clinical consequences include increased work of breathing, ventilator-induced lung injury, an increased mechanical strain on the lungs, and adverse effects on hemodynamics.[3,4] More recent evidence suggests that undiagnosed asynchrony events may result in increased duration of mechanical ventilation and even perhaps an increased mortality rate.[5,6] Identifying patient–ventilator asynchronies is the first step but requires a detailed analysis of ventilator graphical waveforms before being able to correct any discordance. However, waveform interpretation requires a combination of appropriate training and experience. As shown in a recent survey, ventilator waveform interpretation is not yet optimal amongst clinicians.[7] This review will examine the different types of asynchronies and discuss potential solutions.

Types of PVD

The broad term of PVD encompasses several different types of patient–ventilator asynchronies. These asynchronies occur in 3 different phases of the inspiratory cycle: the beginning (trigger), the insufflation phase (flow, rise to pressure), and the transition to exhalation (expiratory cycle). Asynchronies associated with the trigger phase include late trigger (delayed trigger), failed trigger (ineffective effort, missed trigger, wasted effort), false trigger (autotrigger), and reverse trigger. Asynchronies related to the insufflation phase include flow over-assistance and work shifting (under assistance, inadequate flow, flow starvation). Finally, asynchronies that occur at the end of inspiration are early cycle (premature cycle) and late cycle (delayed cycle). Figure 1 illustrates the various asynchronies and where they occur in the respiratory cycle. Ventilator-related dyspnea is an uncomfortable sensation that can arise from PVD. However, in many cases, dyspnea is difficult to identify if the patient is on the ventilator and uncommunicative.

Asynchronies That Occur During the Trigger Phase (Trigger Asynchrony)

Trigger asynchrony is a mismatch between the patient's neural inspiratory effort and the initiation of a ventilator-delivered breath. This type of asynchrony manifests as a late trigger, failed trigger, false trigger, and reverse trigger.

Late Trigger/Failed Trigger

Clinically a late trigger can be observed when there exists a time gap between the onset of the patient's inspiratory effort and the beginning of the mechanical ventilator breath. In the event of a failed trigger, there will be a patient effort to trigger with no response from the ventilator. Figure 2 represents failed triggers, which can be identified through waveform interpretation, particularly in the flow and pressure tracings. Careful clinical assessment involves palpating the patient's upper abdomen to detect the patient's trigger effort while observing the ventilator graphics to see when (or if) the ventilator starts breath delivery. Causes of a late trigger

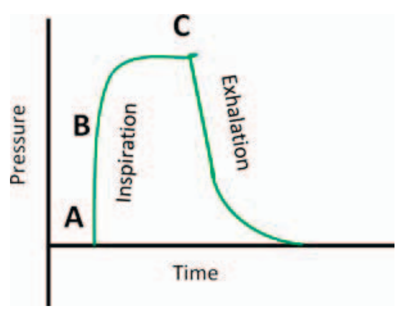

Fig. 1. Types of patient–ventilator asynchronies. **(A)** Asynchronies at the start of inspiration; Late trigger, Failed trigger, False trigger, Reverse trigger. **(B)** Asynchronies during insufflation; Flow over assistance, Work shifting. **(C)**. Asynchronies at the end of inspiration (transition to exhalation); Early cycle, Late cycle.

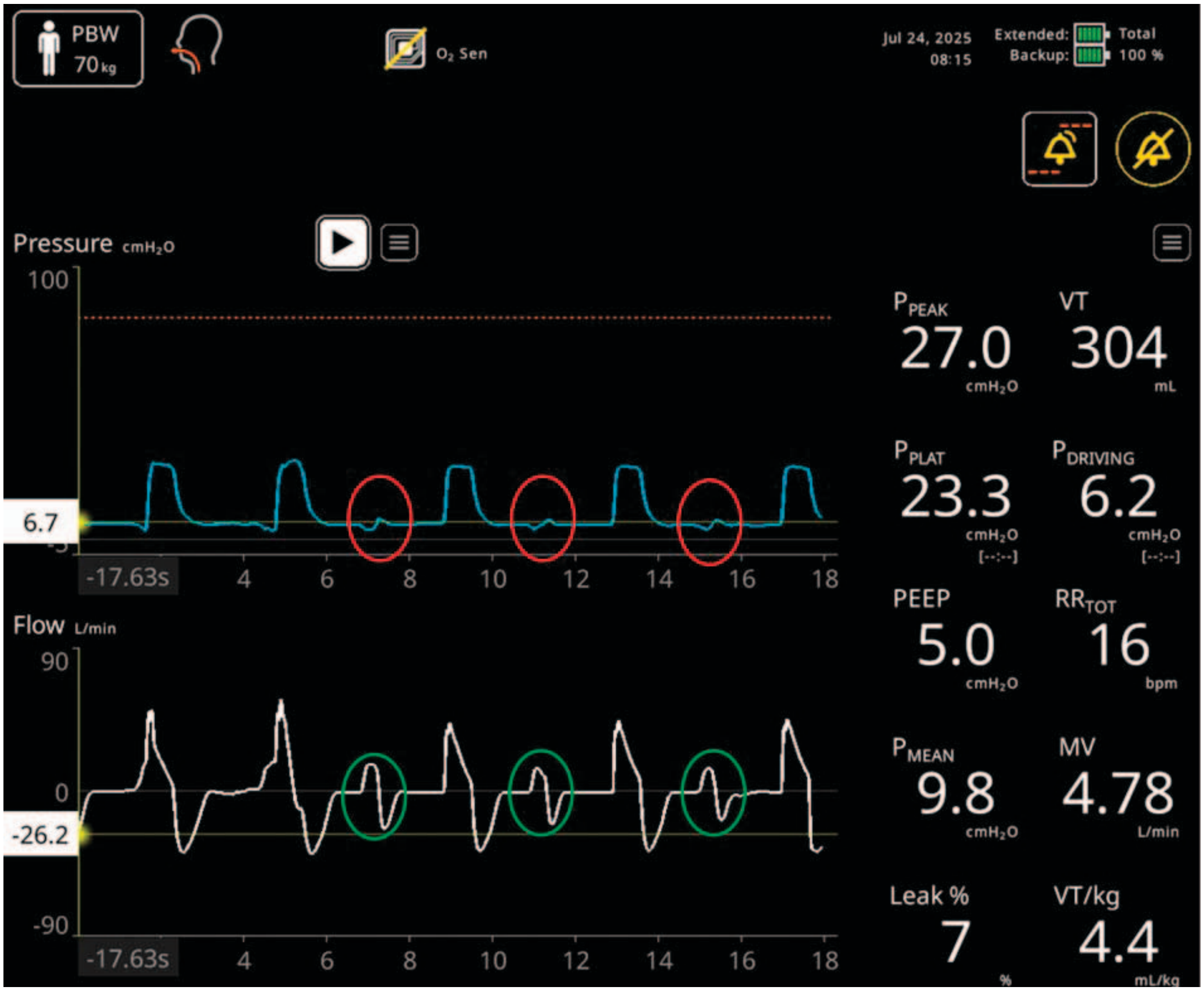

Fig. 2. Failed triggers. The circles represent failed triggers in the flow (flow reversals—green) and pressure (pressure dips—red) waveforms. (Modified with permission from Nihon Kohden).

and failed trigger include a trigger sensitivity setting with a high trigger threshold (too insensitive), over support from the previous breath (high inspiratory pressure levels and tidal volumes), and the presence of intrinsic PEEP (leading to dynamic hyperinflation). Late triggers and failed triggers result in an increased respiratory work of breathing, diaphragmatic fatigue, and patient distress, possibly leading to increased sedation usage.

It is important to identify and correct failed triggering because the ongoing presence of failed triggering has been associated with prolonged duration of mechanical ventilation, extended ICU stay, and higher in-hospital mortality.[8,9]

Correcting late and failed triggering may range from a simple ventilator adjustment to more complex strategies. The trigger sensitivity is a good place to start. If the trigger is set inappropriately high, examination of the patient will reveal extra work to initiate a ventilator breath or effort from the patient with no response from the ventilator. In this scenario a simple adjustment to reduce the trigger threshold may be all that it takes to eliminate late and failed triggers while taking care to avoid false triggers (see below). Assessing the patient's work to trigger the ventilator should be part of every monitoring visit by the respiratory therapist, as clinical situations are subject to change.

The presence of intrinsic PEEP and resulting dynamic hyperinflation pose a different problem for clinicians. Factors that may contribute to the formation of intrinsic PEEP include prolongation of the inspiratory time constant (over support from the previous breath—large tidal volumes and/or inspiratory pressures), prolongation of the expiratory time constant (airway obstruction, high compliance), or a breathing frequency that does not allow enough time for complete exhalation. Solutions for late and failed triggers that result from over support involve adjusting the ventilator settings to reduce the inspiratory pressure or tidal volume. However, clinicians need to be aware that these adjustments may result in a reduction of minute ventilation that may adversely affect the Pa_{CO_2} and pH, as well as increase dyspnea.

Newer technologies that may be helpful with triggering problems include neurally-adjusted ventilatory assist (NAVA) and automated algorithms. NAVA uses electromyographic signals from the diaphragm (EA_{di}) detected via a specialized nasogastric feeding catheter, which is attached to a Servo u ventilator (Getinge, Solna, Sweden). NAVA essentially measures neural respiratory drive (via the diaphragm) as a means of controlling the delivery of inspiratory support by a mechanical ventilator. There is some evidence to suggest benefit in terms of patient–ventilator synchrony, however more research is needed.[10] Automated trigger and cycle algorithms are designed to optimize patient–ventilator synchrony during the trigger and cycle phases of a breath. An example is IntelliSynch+ (Hamilton Medical, Bonaduz, Switzerland), which is incorporated within the Hamilton ventilators. It uses a proprietary algorithm to enhance both breath initiation and breath termination. IntelliSynch+ has shown promise on the bench in terms of synchrony, but again, more research is needed.[11]

Late and failed triggers from the presence of intrinsic PEEP because of a disease condition are more difficult to correct. If appropriate clinical assessment of the patient is not done, late and failed triggers may go unnoticed. Using esophageal manometry greatly enhances a clinician's ability to recognize late triggering and failed triggers. However, esophageal manometry is somewhat invasive and not commonly done. A representation of issues created by the presence of intrinsic PEEP is illustrated in Figure 3. An abnormally high pressure gradient is created that the patient must overcome to trigger a breath. Solutions for late and failed triggers in this scenario include strategies aimed at reducing intrinsic PEEP and perhaps adding extrinsic PEEP to reduce the intrinsic PEEP to circuit pressure gradient. Strategies aimed at reducing the intrinsic PEEP should start with treating the underlying cause. In the case of obstructive airway disease, the use of bronchodilators and corticosteroids may help relieve the intrinsic PEEP. From the ventilator perspective, increasing the time at exhalation may promote more complete exhalation. The usual methods are to decrease inspiratory time and to decrease the breathing frequency. Both of these methods, however, have potential consequences. If the inspiratory time is shortened too much, double triggering (early cycling) may result because the ventilator inspiratory time is too short for the patient's neural inspiratory time. Decreasing the breathing frequency will drop the minute ventilation and may adversely affect the Pa_{CO_2} and pH. In pressure-targeted ventilation reducing the level of support (inspiratory pressure) may aid in more complete exhalation. This strategy will likely decrease the minute ventilation, resulting in alterations of the Pa_{CO_2}, pH and level of dyspnea. If the intrinsic PEEP cannot be eliminated and the patient is having trouble triggering the ventilator, adding extrinsic PEEP could reduce the trigger work load. By moving the circuit pressure closer to the intrinsic PEEP level, the patient does not have to work as hard to overcome the smaller pressure gradient. If successful, the patient's work of triggering decreases and comfort is improved. Clinicians, however, must be careful not to titrate the PEEP too high, as this would add to the work of exhalation and may increase the intrinsic PEEP.

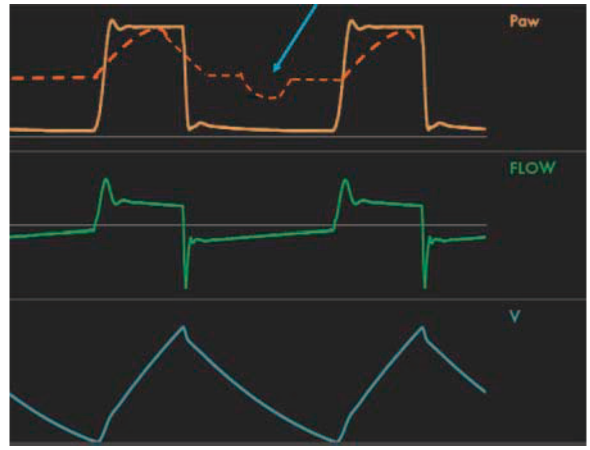

Fig. 3. Graphic representation of a failed trigger because of the presence of intrinsic PEEP. The dotted line represents the alveolar pressure of the patient. The blue arrow points to an effort to trigger a breath from the ventilator. In this case the patient is unable to overcome the intrinsic PEEP and reach the sensitivity threshold, and a failed trigger results.

False Trigger

A false trigger occurs when the ventilator incorrectly identifies a signal to initiate inspiration and responds with an unwanted breath. False triggering can be difficult to pick up from the ventilator waveforms because of the advanced sensitivity of modern-day ventilators. Often an arterial blood gas showing a respiratory alkalosis will be the first hint of false triggering occurring. Things that can cause this include (1) a sensitivity setting that is too sensitive, (2) a circuit leak, (3) vigorous cardiac oscillations, and (4) fluid in the ventilator circuit (particularly in the expiratory limb). Once again, checking the sensitivity setting is a good place to start. A circuit leak can cause the circuit pressure to dip or the bias flow to be affected to the point that the ventilator perceives a patient effort. If there is a circuit leak, this will be visible in the volume tracing which doesn't return to baseline on exhalation. If a circuit leak is suspected, then "walking" the circuit can take place to find the source. Vigorous cardiac oscillations and fluid in the circuit can produce pressure swings that fool the ventilator into thinking a patient effort has occurred. A readjustment of the sensitivity setting may be necessary in the case of cardiac oscillations. If fluid is present in the circuit, it should be drained.

Reverse Triggering

Reverse triggering is a phenomenon that occurs when a time-triggered mechanical breath, followed by evidence of patient effort that may or may not result in a separate, subsequent mechanically assisted breath.[12–14] Because the ventilator "triggers" the patient to breathe, a reversal of the normal series of events in mechanical ventilation, it has thus been termed "reverse triggering."[12] Although other forms of patient–ventilator asynchrony usually represent harmful *interactions* between the patient and the ventilator, reverse triggering is better conceptualized as an abnormal patient *reaction* to a stimulus generated by the ventilator.

Physiology of reverse triggering

The physiologic cause of reverse triggering is unknown. One prevailing theory is that reverse triggering represents respiratory entrainment.[12,14,15] Entrainment occurs when the respiratory central pattern generator within the brainstem generates patient inspiratory efforts in coordination with a repetitive, regular external stimulus.[14–16] For example, during cycling, the central pattern generator will adjust a person's breathing pattern to coincide with the frequency of limb movement.[17] Similar responses to stimuli such as walking on a treadmill, speaking, and singing have been observed.[18–20] So, as a ventilator with a set breathing frequency provides a regularly recurring external stimulus in the form of a ventilated breath, it is thought that this can entrain subsequent patient-triggered breaths. However, it must be noted that although there is some preliminary preclinical evidence for entrainment as the cause of reverse triggering, it is not definitive.

An alternative explanation is that reverse triggering represents the "unmasking" of a spinal reflex. As reverse triggering is most frequently observed in patients who are deeply sedated, are transitioning from deep sedation to lighter sedation, or have primarily neurologic injuries, it has been postulated that alterations in higher level cortical function by sedation and/or injury allow for spinal reflexes that have previously been suppressed. Observations of reverse triggering in patients with brain death or without brainstem reflexes have been used as evidence to support the assertion of a spinal reflex as the cause of reverse triggering, but as with the entrainment theory, this is still controversial.[21]

Clinical presentation

Clinically, reverse triggering closely resembles other forms of ventilator asynchrony that cause "breath-stacking" or "double triggering."[22] Thus, clinicians must maintain a high index of suspicion for reverse triggering in order to delineate it from other common asynchronies.[23] As mentioned above, reverse triggering is usually observed in patients who are deeply sedated, are transitioning from deep sedation to lighter levels of sedation, or have a primarily neurologic injury, so these patients should be closely monitored for signs of reverse triggering.[13,14,21,24]

The hallmark of reverse triggering is the occurrence of a patient trigger effort immediately following a mandatory ventilator breath, which may be visible on bedside ventilator graphic analysis (Fig. 4). A patient trigger effort immediately following a previous patient-triggered breath (may be seen as a negative deflection in the pressure waveform during a volume-controlled breath) is likely not reverse triggering and instead may represent other forms of asynchrony such as flow asynchrony or cycle asynchrony.[22,23]

The frequency of reverse triggering may vary greatly, with some patients having a reverse-triggered breath following all or nearly all ventilator-triggered breath, or only a fraction of ventilator-delivered breaths being followed by reverse-triggered breath.[25] The frequency may also change over time as exposure of an individual patient to sedatives and analgesics changes.[26,27]

Clinical importance

As reverse triggering has only recently been recognized as a form of asynchrony, the full extent of its clinical importance is not yet known. There is obvious concern that the "double-stacking" of breaths may cause excessively high tidal volumes and transpulmonary pressures that could lead to ventilator-induced lung injury.[28,29]

In addition, the presence of an exuberant respiratory effort because of reverse triggering has been associated with diaphragmatic damage in a preclinical model. Conversely, the activation of a *mild* diaphragm contraction may prevent muscle disuse atrophy of the diaphragm in the same preclinical models.[29] However, robust clinical data to support the findings of a potentially protective effect of reverse triggering are lacking.

Management of Reverse Triggering. Management of reverse triggering is based on experiential reports and conceptual physiologic models. Because of the association of reverse triggering with deeper levels of sedation, reducing patient sedation levels is often recommended if safe.[30] If it is unsafe or not possible to lighten sedation, then neuromuscular blockade may be considered to abolish reverse triggering.[31] Changes to the ventilator may also be considered, such as transitioning to support mode of ventilation such as pressure support, though there are some

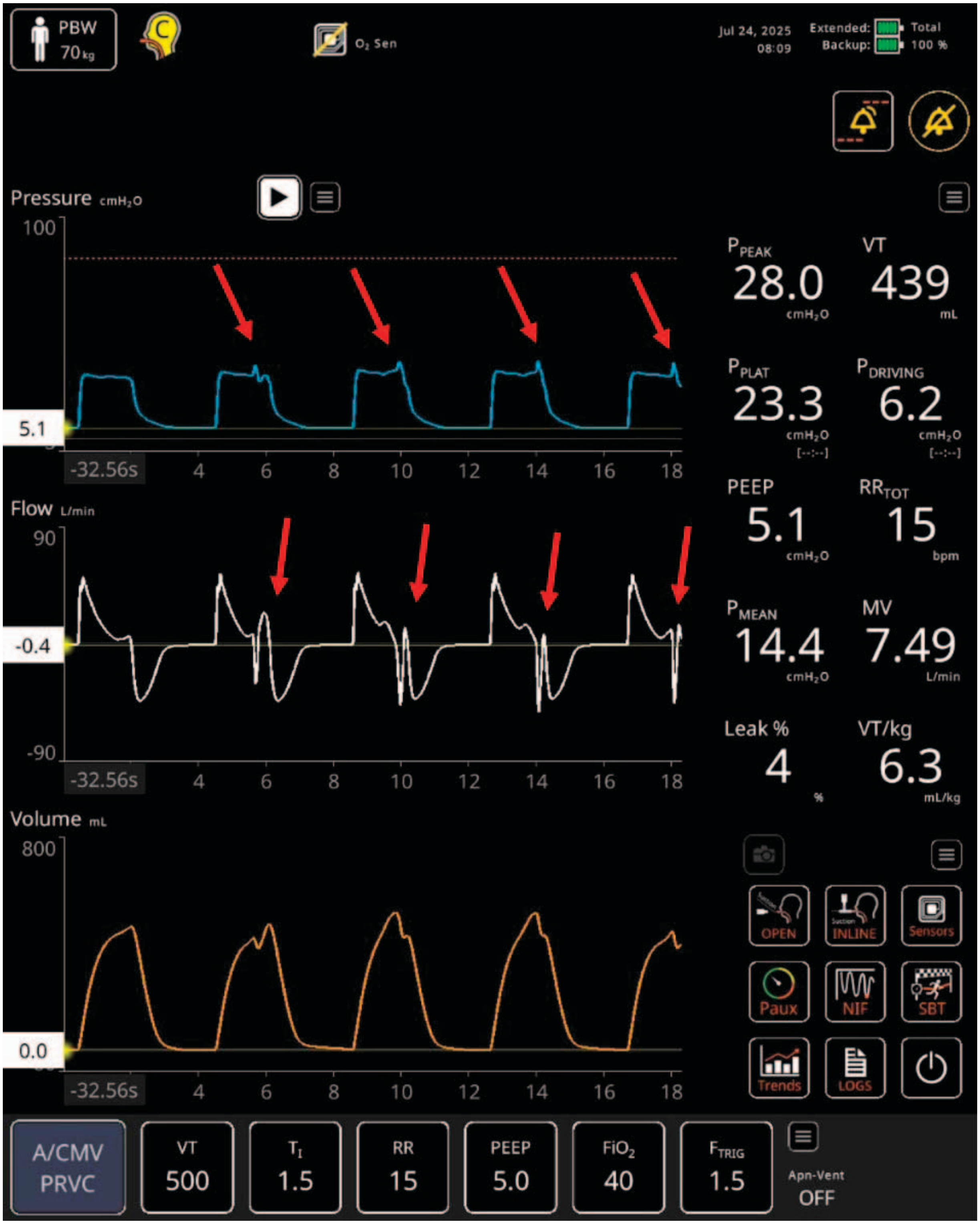

Fig. 4. Reverse triggering is present. During the mandatory time-triggered ventilator breaths, patient trigger efforts can be seen in the pressure tracing (top waveform—red arrows) and the flow tracing (middle waveform—red arrows). (Modified with permission from Nihon Kohden).

reports of reverse triggering in patients managed with support modes of ventilation.[27,32,33]

In addition, as entrainment is most often cited as the physiologic basis for reverse triggering, lowering the set breathing frequency delivered by the ventilator has been hypothesized to diminish or even abolish the entrainment of the central pattern generator by changing the regularity and frequency with which the patient is exposed to the external stimuli.[34] Finally, both reducing sedation and decreasing breathing frequency may allow for the patient to set their own breathing frequency entirely, eliminating the stimuli for reverse triggering entirely. Other changes to the ventilator, such as adjusting tidal volume, PEEP, and transitioning from flow-limited to pressure-limited modes of ventilation have been proposed, but the effect of these interventions on reducing the frequency of reverse triggering is unclear.[15,30,35] A reasonable approach would be to try and reduce sedation and breathing frequency if clinically tolerated, consider earlier use of a support mode of ventilation such as pressure support, and if those interventions are not tolerated and/or ineffective, use neuromuscular blockade if reverse triggering is felt to be increasing a patient's risk of ventilator-induced lung injury.

Asynchronies During Insufflation (Flow Asynchrony)

Flow over assistance

Flow over assistance is a situation where the ventilator provides more flow during insufflation than the patient can tolerate. Although rare, this over support can occur in both volume (set flow too rapid) and pressure ventilation (high inspiratory pressure setting) and can be seen on the pressure waveform as a spike at the beginning of the pressure tracing. Flow over assistance can cause patient discomfort and may alter the cycling off to expiration in pressure support by creating a new exaggeratedly high peak flow. In volume ventilation the flow can be adjusted either directly or indirectly (the inspiratory time and tidal volume will determine the flow on some ventilators). So, adjusting the set flow should alleviate this. In pressure ventilation, flow over support can result from either an excessive inspiratory pressure setting or a short rise time. In pressure support, this phenomenon can lead to an early cycle since the flow decay is likely to be more rapid. Again, these are settings that can be adjusted directly by clinicians to prevent this over support (reducing the inspiratory pressure level or increasing the rise time). It should be noted that decreasing the set inspiratory pressure is likely to reduce the delivered tidal volume and may affect the Pa_{CO_2} and pH.

Flow Under Assistance (Inadequate Flow, Flow Starvation, Work Shifting)

Flow under assistance is a mismatch where the patient's innate flow demand exceeds the ventilator flow response. It occurs more frequently in volume ventilation, where the flow is set and cannot be modified by the patient. This can be diagnosed from the pressure tracing on the ventilator. There is a pressure waveform deformation where the pressure waveform exhibits a downward concavity during inspiration (Fig. 5). The total pressure required for a ventilator breath is a combination of what the ventilator provides (P_{vent}) and what the patient contributes (P_{mus}). Under-assistance where inadequate flow is being delivered from the ventilator has also been

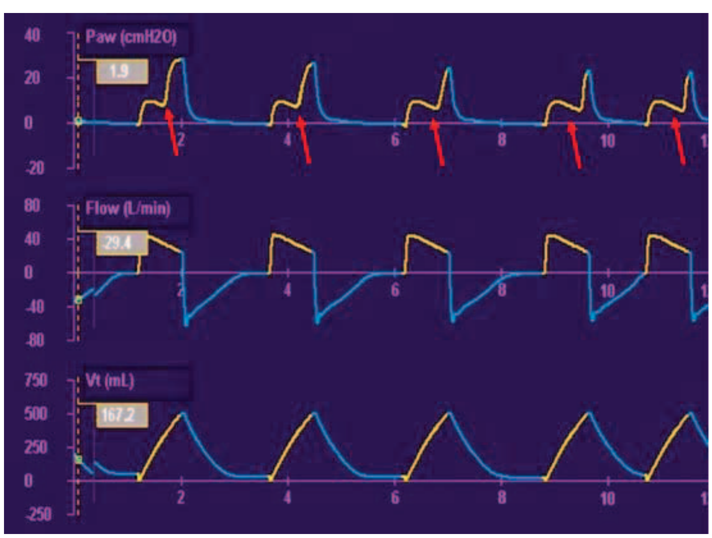

Fig. 5. Work shifting (inadequate flow) in volume ventilation Note the downward "pulling" of the pressure tracing (red arrows). Since flow is set, the patient's efforts to get more flow are reflected in the pressure waveform.

referred to as "work shifting," in which the ventilator fails to unload the respiratory muscles adequately, and the work shifts to the patient. This may be associated with increased dyspnea, patient self-induced lung injury (high pleural pressure swings leading to high transpulmonary pressures), and myotrauma.[36–38] In volume ventilation, the flow can be directly adjusted to meet the patient's flow demands. Unfortunately, turning up the flow has the unintended consequence of reducing the inspiratory time, and it can be difficult to match the patient's flow demands on every breath because of their intrinsic variability. If the inspiratory time gets shortened dramatically from the increase in flow, double triggering (early cycling) may result because of the mismatch of ventilator inspiratory time and the patient's neural inspiratory time. Changing to a pressure mode may be the best choice to alleviate flow under support. In pressure ventilation the flow is variable and adjusted by the ventilator to match patient effort. A concerning consequence of the increased flow is a corresponding increase in tidal volume. (Figs. 6 and 7 illustrate both solutions to flow under-assistance). Although patient comfort should improve, care must be taken to ensure safe tidal volumes. Although rare, flow under assistance can occur in pressure ventilation if the rise time is set too long. If the ventilator takes too long to reach the set pressure, the patient may become uncomfortable. Fortunately, the rise time is a direct setting on the ventilator, so a clinician can solve flow starvation here by merely adjusting the rise time and assessing patient effort.

Asynchronies at the End of Inspiration (Cycle Asynchrony)

Early and late cycling are asynchronies in which the ventilator does not end the breath when the patient desires it to end.

Early cycling

If the ventilator ends the inspiration before the patient wants it, it is termed early cycling (also known as premature or short cycling). The patient's inspiratory effort continues on through the early part of the ventilator exhalation. If the mismatch is severe, double

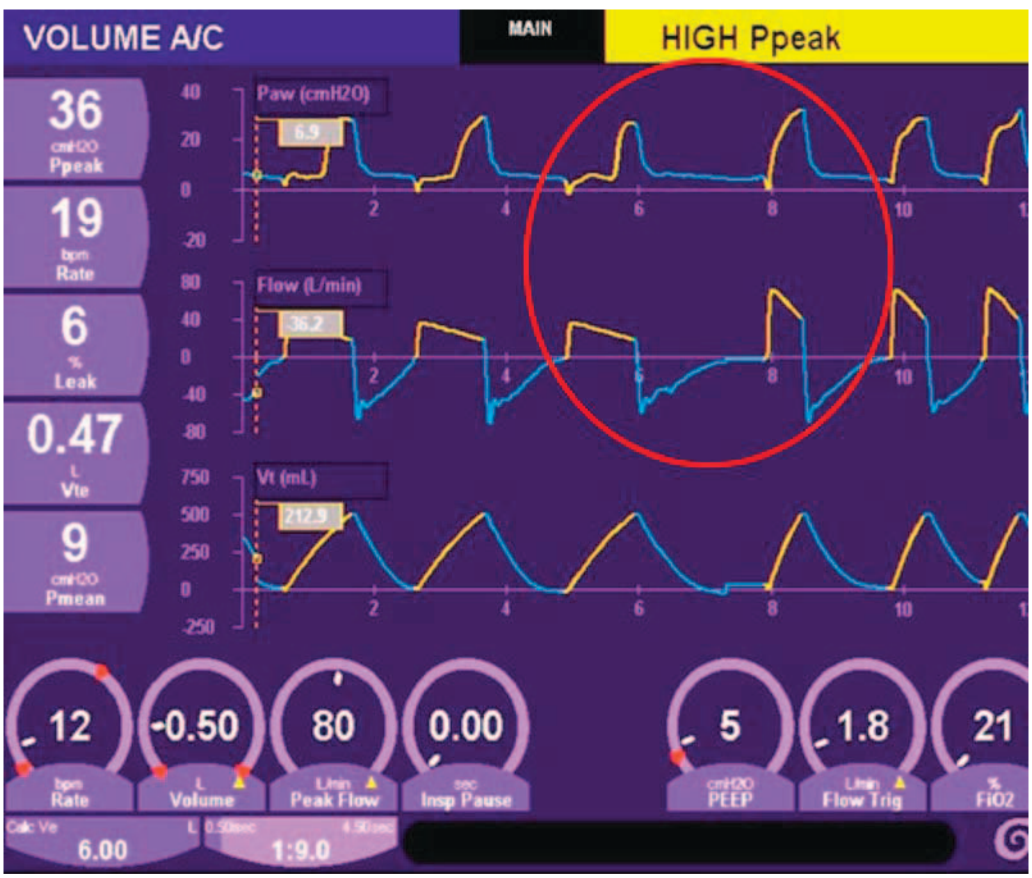

Fig. 6. Work shifting (inadequate flow) corrected by increasing set flow in volume ventilation. Note the increase in flow in the last 3 breaths has resulted in a significantly shorter inspiratory time.

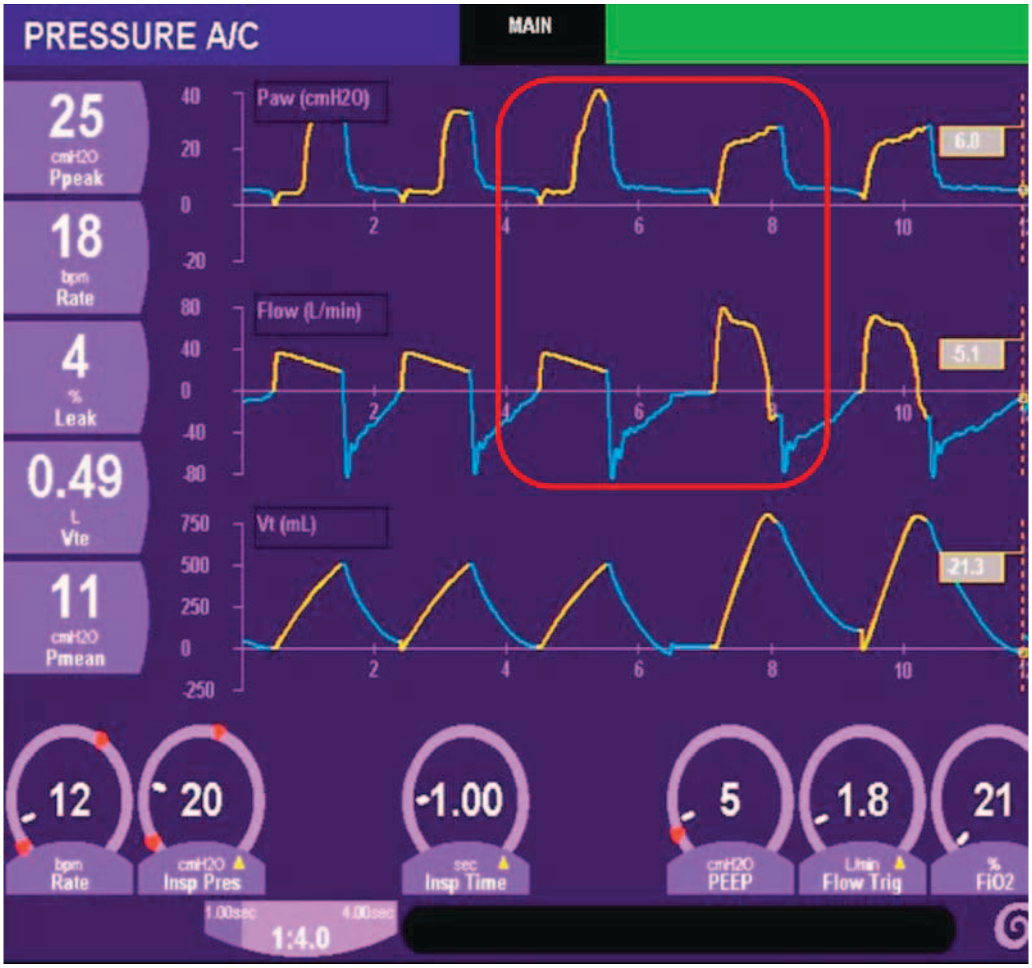

Fig. 7. Work shifting (inadequate flow) corrected by switching to pressure ventilation. The first 3 breaths are volume breaths exhibiting flow starvation. The next 2 breaths are pressure breaths with a resulting higher flow and higher tidal volume.

triggering (often called breath-stacking) may occur. Double triggering can produce high tidal volumes outside what is considered safe. The ventilator does not recognize that the patient effort is merely a continuation of the previous breath; it interprets the continued effort as a new signal to initiate a new breath. Early cycling can occur in any mode where there is patient effort. Figure 8 shows some double-triggered breaths. Potential remedies are: (1) In volume ventilation, increase inspiratory time by either increasing tidal volume (if safe to do so) or reducing inspiratory flow (making sure flow starvation does not result). Some makes of ventilators do have an inspiratory time setting that can be adjusted, but this also affects inspiratory flow. (2) In pressure control, there is a direct inspiratory time setting that can be lengthened. (3) In pressure support, decreasing the cycle threshold (eg, 40–25%) will help prolong the inspiratory time.

Late cycling

Late cycling occurs when the ventilator inspiratory time exceeds the patient's neural inspiratory time. Late cycling can be seen when the patient recruits respiratory muscles to try and exhale while the ventilator maintains inspiration (Fig. 9). Similar to early cycling, late cycling can occur in any mode. Potential remedies are (1) in volume ventilation, decrease inspiratory time directly or indirectly by decreasing tidal volume and/or increasing the flow setting; (2) in pressure control, the inspiratory time is a direct setting that can be reduced to match patient effort; (3) in pressure support, increase the flow threshold to a higher number. In pressure support, the flow threshold is especially important for a COPD patient. Patients with COPD have highly compliant lungs and some loss of elastic recoil. This results in them having a slow flow decay, which may take longer to reach the flow threshold. Adjusting the flow threshold to a higher number in this

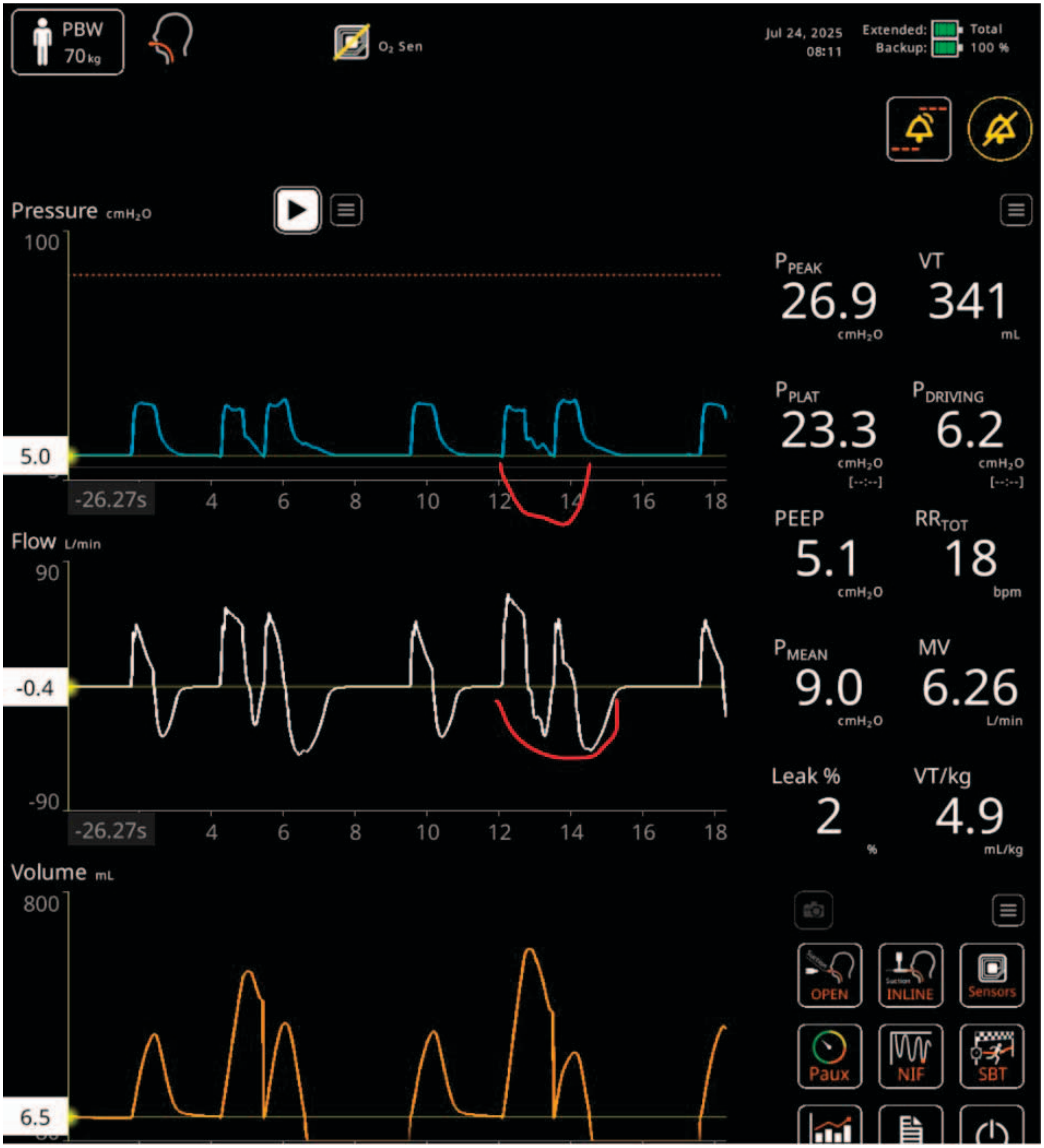

Fig. 8. Double triggering resulting from early cycling. The red lines represent patient effort that carries on past the end of the first ventilator breath (modified with permission from Nihon Kohden).

patient population is especially important in maintaining a normal inspiratory time.

Summary

Patient–ventilator asynchrony can occur at the beginning of a breath, during the insufflation phase, and at the termination of a breath. Table 1 depicts the various asynchronies, their causes, and potential solutions to optimize synchrony. Correction of patient–ventilator asynchronies involves several steps that should be followed. Recognition of the asynchrony is the first step. This involves ventilator waveform analysis coupled with the clinical assessment of the patient. Next, the appropriate changes should be made on the ventilator. Finally, ongoing assessment is needed to ensure that optimal synchrony is maintained. Table 2 illustrates an algorithm that could be used to correct asynchronies.

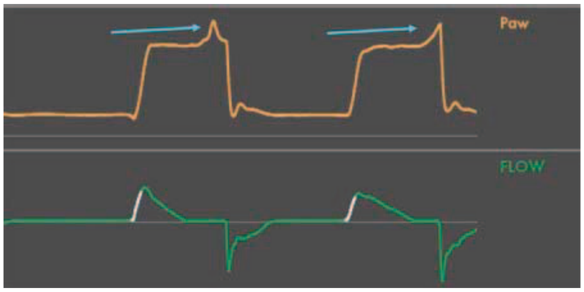

Fig. 9. Late cycling. Note the pressure spikes near the end of each breath (blue arrows)—patient efforts to start exhalation.

Dyspnea

Dyspnea can be defined in several ways, including an uncomfortable sensation associated with breathing or a feeling of not getting enough air, being short of breath, feeling breathless, or experiencing "air hunger." Whatever word or phrase is used to describe dyspnea, this sensation occurs when the drive to breathe cannot be easily or comfortably satisfied with the body's ability to move air. This drive is multifaceted and can be mediated or even magnified by individual or overlapping chemical, mechanical, or psychological factors present in the patient. Chemical stimulants of respiratory drive include hypercapnia, acidosis, and/or hypoxemia. Pulmonary stretch receptors, as well as chest wall mechanoreceptors and proprioceptors, provide mechanical feedback to signal an increased work of breathing.[39] These mechanical stimuli are further amplified in the setting of limited tidal volume, which has become a standard of care in the management of patients requiring invasive mechanical ventilation.[40] Finally, psychological factors—such as anxiety, panic attacks, and voluntary hyperventilation—can increase respiratory drive either independently of chemical or mechanical stimuli or in response to them.

The control of ventilation is located in the brainstem and typically adjusts seamlessly to subtle chemical or mechanical

Table 1. Patient–ventilator asynchronies, causes and potential solutions

Asynchrony	*Description*	*Causes*	*Correction*
Trigger asynchrony			
Late trigger (delayed trigger)	• Time delay between patient effort and ventilator activation	• Trigger threshold set too high • Presence of auto-PEEP	• Adjust sensitivity • Check for over support from the previous breath • Add extrinsic PEEP to reduce triggering WOB
Failed trigger (ineffective effort, missed trigger, wasted effort)	• Patient effort fails to trigger the ventilator	• Trigger threshold set too high • Presence of auto-PEEP	• Adjust sensitivity • Check for over support from the previous breath • Add extrinsic PEEP to reduce triggering WOB
False trigger (auto trigger)	• Breath initiated by something other than patient effort	• Trigger threshold set too low • Vigorous cardiac oscillations • Water sloshing in the circuit	• Adjust sensitivity • Adress cardiac oscillations, if possible • Drain water from the circuit
Reverse trigger	• Patient inspiratory effort after the onset of a timed breath from the ventilator	• Unknown at present	• Paralysis? • Lighten sedation? • Reduce breathing frequency? • PS?
Flow asynchrony			
Flow over support	• Rapid delivery of flow that overwhelms the patient	• VC—flow is set too high • PC/PS—Rise time too fast	• VC—adjust flow setting • PC/PS—adjust rise time
Work shifting (inadequate flow, flow under support)	• Support during breath delivery is not sufficient from the ventilator and the patient has to assume the extra work	• VC—flow is set too low • PC/PS—Rise time set too low, inspiratory set pressure too low	• VC—adjust the flow, switch to PC or PS • PC/PS—adjust the rise time and perhaps the inspiratory pressure
Cycle asynchrony			
Early cycle (premature cycle, short cycle)	• The ventilator terminates the breath before the patient is ready	• VC—V_T set too low, flow is set too high, Ti set too short • PC—Ti set too short • PS - flow termination setting set too high	• VC—increase V_T (as lung-protective ventilation allows), reduce flow, adjust Ti • PC—adjust Ti • PS—lower the flow termination setting
Late cycle	• The ventilator terminates the breath after the patient is ready	• VC—V_T set too high, flow set too low, Ti set too long • PC—Ti set too long • PS—flow termination set too low	• VC—decrease V_T, increase flow, adjust Ti • PC—adjust Ti • PS—increase the flow termination setting

VC, volume control; PC, pressure control; PS, pressure support; V_T, tidal volume; Ti, inspiratory time; WOB, work of breathing.

Table 2. A suggested clinical algorithm for identifying and correcting patient–ventilator asynchrony

Clinical Algorithm for Correction of Patient–Ventilator Asynchrony
Recognition: • Increased respiratory distress • Tachypnea • Visible discordance between patient effort and ventilator response Assessment: • Waveform ventilator analysis combined with corresponding patient assessment Diagnosis: • Determine the underlying causes of the identified asynchrony o Evaluate ventilator settings o Detection of auto-PEEP Management: • Optimize ventilator settings • Sedation only as a last resort (increased respiratory drive) Reassessment: • Monitor the patient's response to ensure the asynchrony has been resolved

changes during daily activities. Dyspnea occurs when these changes become difficult to compensate and arises when factors that stimulate respiratory drive activate the corticolimbic regions of the insular cortex—a key area involved in the perception of both pain and breathlessness.[39,41] In essence, when the demand for ventilation is not adequately met, the brain begins screaming "BREATHE!," which is perceived subjectively as dyspnea.

Patients receiving mechanical ventilation often struggle to communicate their needs because of factors including physical weakness, inability to speak, sedation, and delirium. They may lack full control over their own breathing as clinicians prioritize safe ventilator settings, such as low tidal volumes, and their underlying illnesses may further impair their ability to breathe comfortably. These combined factors contribute significantly to the experience of dyspnea in mechanically ventilated patients. Moderate to severe dyspnea is reported in > 40% of patients who are able to communicate while on mechanical ventilation.[42] Although assessing dyspnea in noncommunicative patients is more challenging, measures of increased work of breathing can be utilized to assist in identification. Fortunately, the identification and subsequent management of dyspnea in mechanically ventilated patients is receiving more attention with recent guidance for clinicians published by the European Respiratory Society and the European Society of Intensive Care Medicine.[43]

The combination of difficulty in communicating needs, dyspnea, and delirium in critically ill patients requiring mechanical ventilation leads to significant anxiety and posttraumatic stress disorder (PTSD) that persist beyond the acute critical illness phase.[44,45] Measures to improve communication, reduce delirium, and mitigate dyspnea can improve anxiety and PTSD in this population.[46]

Dyspnea in mechanical ventilation

Many of the physiologic changes that are present with dyspnea occur whether one is healthy and performing strenuous exercise or critically ill and requiring mechanical ventilation. The healthy runner becomes tachypneic, tachycardic, and uses accessory muscles to breathe and has an increase in systemic blood pressure. All of this is associated with ensuring adequate minute ventilation and cardiac output are achieved to deliver needed oxygen to working tissues and remove the CO_2 being produced during strenuous exertion.

Eventually, the runner reaches a physiological threshold beyond which continued exertion is unsustainable. Upon cessation of activity, both breathing frequency and heart rate rapidly return toward baseline levels as oxygen consumption and carbon dioxide production decline. The runner does not truly hit an anxiety-provoking level of dyspnea because the termination of the activity driving the dyspnea is under their control. Whereas a critically ill patient on a ventilator may be living in a constant state of dyspnea that they cannot simply resolve because they can't control the activity driving the dyspnea.

As the sensation of dyspnea intensifies, the respiratory drive correspondingly increases. In addition to a physical exam demonstrating accessory muscle use, tachypnea, tachycardia, and increased work of breathing, the ventilator can also provide clues to the presence of increased and potentially unmet respiratory drive. This may be identified by several physiologic indicators, including an increased use of accessory inspiratory muscles, a high early inspiratory flow in patients receiving pressure-regulated modes of ventilation, a drop in airway pressure during inspiration in volume-regulated modes, an elevated airway occlusion pressure ($P_{0.1}$), a pronounced negative deflection in esophageal pressure (if an esophageal balloon is in place), or an increased electrical activity of the diaphragm.[47,48] Observational scales, such as the mechanical ventilation-respiratory distress observation scale (MV-RDOS) or the intensive care RDOS (IC-RDOS), provide an additional objective tool for clinicians to utilize when trying to identify dyspnea.[49,50] Worsened dyspnea and subsequent increased drive to breathe in a patient receiving invasive mechanical ventilation can indicate worsening of an underlying condition, a new condition, or a problem with the patient–ventilator interaction, and it is associated with higher mortality.[51]

The sensation of dyspnea is only worsened by persistent asynchrony, which can lead to an escalating drive to breathe, inefficient work of breathing, and ultimayely, if not effectively addressed, respiratory muscle fatigue and diaphragmatic injury.[52,53]

It stands to reason that dyspnea is at least partially related to the degree of patient–ventilator asynchrony. Trigger, flow, and cycle asynchronies can all contribute to the sensation of dyspnea as the patient tends to "fight" with the ventilator. It is not clear which type of asynchrony contributes to dyspnea the most, although one study showed that many of their subjects cited inadequate flow in volume assist control as the main source of their dyspnea.[45] All types of asynchrony probably contribute to the feeling of dyspnea, and it makes sense to ensure the ventilator is set in such a way as to optimize patient–ventilator synchrony.

Treatment of dyspnea in mechanical ventilation

Treatment of dyspnea in patients receiving mechanical ventilation first involves addressing the underlying disease process. Whether the driver is chemical, mechanical, or psychological, treatment of this underlying process can reduce the need for higher minute ventilation, drive to breathe, and dyspnea.

Concurrently, although the underlying condition causing or contributing to an increased drive to breathe is being addressed, attention must be directed to the mechanical ventilator. The need for increased minute ventilation should be matched in the most comfortable and efficient manner while still providing for lung-protective ventilation. Patient–ventilator asynchrony hinders the comfort and efficiency of mechanical ventilation and contributes to dyspnea and anxiety. In fact, patients receiving mechanical ventilation who report dyspnea overwhelmingly describe that they are not getting enough air rather than feeling as though they are working too hard to breathe, and dyspnea is significantly worse in those with lower tidal volumes independent of lung mechanics.[54] Simply adjusting ventilator setting to improve patient–ventilator interactions can reduce dyspnea in 35–50% of patients.[45,49]

Due to the overlapping neural location of the perception of pain and dyspnea, both endogenous and exogenous opioids can modulate the perception of dyspnea, particularly in patients whose symptoms persist despite treatment of the underlying condition and/or optimization of patient–ventilator interaction.[39] Patients frequently receive both sedatives and opioids while receiving mechanical ventilation. However, opioids alone can help mitigate dyspnea in the majority of patients not responsive to ventilator adjustments alone.[46] Opioid dosing can reduce overall patient–ventilator asynchronies without worsening sedation levels into morbid ranges.[55]

Dyspnea is common in mechanically ventilated patients, is highly associated with anxiety and pain, and is improved in many patients by altering the ventilator settings to improve patient–ventilator interactions.[56] Effectively identifying dyspnea and increased work of breathing is paramount to mitigating the long-term complications associated with persistent dyspnea in patients receiving mechanical ventilation.

Summary

Patient-ventilator discordance is a common phenomenon that is not completely understood. It presents a real challenge for clinicians at the bedside. In many cases, patient-ventilator discordance goes unnoticed and could lead to diaphragmatic fatigue and overuse of sedation. These, in turn, can lead to unwanted clinical outcomes such as a longer duration of mechanical ventilation (and its associated risks) and perhaps a higher mortality risk. Bedside patient assessment and vigilant monitoring of ventilator waveforms remain the cornerstones of identifying and correcting asynchronous events.

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References

1. De Wit M, Pedram S, Best AM, et al. Observational study of patient-ventilator asynchrony and relationship to sedation level. J Crit Care 2009;24(1):74–80; doi: 10.1016/j.jcrc.2008.08.011
2. Thille AW, Rodriguez P, Cabello B, et al. Patient-ventilator asynchrony during assisted mechanical ventilation. Intensive Care Med 2006;32(10): 1515–1522; doi: 10.1007/s00134-006-0301-8
3. Beitler JR, Sands SA, Loring SH, et al. Quantifying unintended exposure to high tidal volumes from breath stacking dyssynchrony in ARDS: the BREATHE criteria. Intensive Care Med 2016;42(9):1427–1436; doi: 10.1007/s00134-016-4423-3
4. De Haro C, Ochagavia A, Lopez-Aguilar J, et al.; Asynchronies in the Intensive Care Unit (ASYNICU) Group. Patient-ventilator asynchronies during mechanical ventilation: current knowledge and research priorities. Intensive Care Med Exp 2019;7(Suppl 1):43; doi: 10.1186/s40635-019-0234-5
5. Subira C, de Haro C, Magrans R, et al. Minimizing asynchronies in mechanical ventilation: Current and future trends. Respir Care 2018; 63(4):464–478; doi: 10.4187/respcare.05949
6. Blanch L, Villagra A, Sales B, et al. Asynchronies during mechanical ventilation are associated with mortality. Intensive Care Med 2015;41(4): 633–641; doi: 10.1007/s00134-015-3692-6
7. Liu P, Lyu S, Mireles-Cabodevila E, et al. Survey of ventilator waveform interpretation among ICU professionals. Respir Care 2024;69(7):773–781; doi: 10.4187/respcare.11677
8. Vaporidi K, Babalis D, Chytas A, et al. Clusters of ineffective efforts during mechanical ventilation: impact on outcome. Intensive Care Med 2017;43(2):184–191; doi: 10.1007/s00134-016-4593-z
9. De Bie MJ, Rietveld PJ, Ven der Velde-Quist F, et al. The association between patient-ventilator asynchrony and clinical outcomes in mechanically ventilated patients: A systematic review. Crit Care Med 2025;53(11): e2261–e2270; doi: 10.1097/CCM.0000000000006816Epub 2025 Aug 12
10. Mandyam S, Qureshi M, Katamreddy Y, et al. Neurally adjusted ventilatory assist versus pressure support ventilation: A comprehensive review. J Intensive Care Med 2024;39(12):1194–1203; doi: 10.1177/08850666231212807Epub 2023 Nov 15
11. Morinishi K, Itagaki T, Chikata Y, et al. Effects of trigger algorithms on trigger performance and patient-ventilator synchrony. Respir Care 2025; 70(10):1285–1293; doi: 10.1089/respcare.12694
12. Akoumianaki E, Lyazidi A, Rey N, et al. Mechanical ventilation-induced reverse-triggered breaths: a frequently unrecognized form of neuromechanical coupling. Chest 2013;143(4):927–938; doi: 10.1378/chest.12-1817
13. Núñez Silveira JM, Gallardo A, García-Valdés P, et al. Reverse triggering during mechanical ventilation: Diagnosis and clinical implications. Med Intensiva (Engl Ed) 2023;47(11):648–657; doi: 10.1016/j.medine.2023.10.009
14. Rodrigues A, Telias I, Damiani LF, et al. Reverse triggering during controlled ventilation: From physiology to clinical management. Am J Respir Crit Care Med 2023;207(5):533–543; doi: 10.1164/rccm.202208-1477CI

15. de Vries HJ, Jonkman AH, Tuinman PR, et al. Respiratory entrainment and reverse triggering in a mechanically ventilated patient. Ann Am Thorac Soc 2019;16(4):499–505; doi: 10.1513/AnnalsATS.201811-767CC
16. Potts JT, Rybak IA, Paton JF. Respiratory rhythm entrainment by somatic afferent stimulation. J Neurosci 2005;25(8):1965–1978; doi: 10.1523/JNEUROSCI.3881-04.2005
17. Bonsignore MR, Morici G, Abate P, et al. Ventilation and entrainment of breathing during cycling and running in triathletes. Med Sci Sports Exerc 1998;30(2):239–245; doi: 10.1097/00005768-199802000-00011
18. Hill AR, Adams JM, Parker BE, et al. Short-term entrainment of ventilation to the walking cycle in humans. J Appl Physiol (1985) 1988;65(2):570–578; doi: 10.1152/jappl.1988.65.2.570
19. Abbasi O, Kluger DS, Chalas N, et al. Predictive coordination of breathing during intra-personal speaking and listening. iScience 2023;26(8):107281; doi: 10.1016/j.isci.2023.107281eCollection 2023 Aug 18
20. Schmidt MF, Goller F. Breathtaking Songs: Coordinating the neural circuits for breathing and singing. Physiology (Bethesda) 2016;31(6):442–451; doi: 10.1152/physiol.00004.2016
21. Delisle S, Charbonney E, Albert M, et al. Patient-ventilator asynchrony due to reverse triggering occurring in brain-dead patients: Clinical implications and physiological meaning. Am J Respir Crit Care Med 2016;194(9):1166–1168; doi: 10.1164/rccm.201603-0483LE
22. Murias G, de Haro C, Blanch L. Does this ventilated patient have asynchronies? Recognizing reverse triggering and entrainment at the bedside. Intensive Care Med 2016;42(6):1058–1061; doi: 10.1007/s00134-015-4177-3
23. Pham T, Telias I, Piraino T, et al. Asynchrony consequences and management. Crit Care Clin 2018;34(3):325–341; doi: 10.1016/j.ccc.2018.03.008
24. Bourenne J, Guervilly C, Mechati M, et al. Variability of reverse triggering in deeply sedated ARDS patients. Intensive Care Med 2019;45(5):725–726; doi: 10.1007/s00134-018-5500-6
25. Pham T, Montanya J, Telias I, et al.; BEARDS study investigators. Automated detection and quantification of reverse triggering effort under mechanical ventilation. Crit Care 2021;25(1):60; doi: 10.1186/s13054-020-03387-3
26. Luo XY, Zhou YM, Zhou J, et al. Reverse triggering in patients with acute brain injury undergoing controlled ventilation. J Thorac Dis 2024;16(10):6441–6451; doi: 10.21037/jtd-24-694
27. Longhini F, Simonte R, Vaschetto R, et al. Reverse triggered breath during pressure support ventilation and neurally adjusted ventilatory assist at increasing propofol infusion. J Clin Med 2023;12(14):4857; doi: 10.3390/jcm12144857
28. Su HK, Loring SH, Talmor D, et al. Reverse triggering with breath stacking during mechanical ventilation results in large tidal volumes and transpulmonary pressure swings. Intensive Care Med 2019;45(8):1161–1162; doi: 10.1007/s00134-019-05608-y
29. Damiani LF, Engelberts D, Bastia L, et al. Impact of reverse triggering dyssynchrony during lung-protective ventilation on diaphragm function: An experimental model. Am J Respir Crit Care Med 2022;205(6):663–673; doi: 10.1164/rccm.202105-1089OC
30. Murray B, Sikora A, Mock JR, et al. Reverse triggering: An introduction to diagnosis, management, and pharmacologic implications. Front Pharmacol 2022;13:879011; doi: 10.3389/fphar.2022.879011
31. Slutsky AS, Villar J. Early paralytic agents for ARDS? Yes, no, and sometimes. N Engl J Med 20192019;380(21):2061–2063; doi: 10.1056/NEJMe1905627
32. He X, Luo XY, Chen GQ, et al. Detection of reverse triggering in a 55-year-old man under deep sedation and controlled mechanical ventilation. J Thorac Dis 2018;10(9):Ee682–Ee685; doi: 10.21037/jtd.2018.08.09
33. Lassola S, Cucino A, Bellani G. Is reverse triggering in pressure-support ventilation an oxymoron? Maybe not!. Am J Respir Crit Care Med 2025;211(3):e5–e6; doi: 10.1164/rccm.202404-0766IM
34. Mellado Artigas R, Damiani LF, Piraino T, et al. Reverse riggering dyssynchrony 24 h after initiation of mechanical ventilation. Anesthesiology 2021;134(5):760–769; doi: 10.1097/ALN.0000000000003726
35. Baedorf Kassis E, Su HK, Graham AR, et al. Reverse trigger phenotypes in acute respiratory distress syndrome. Am J Respir Crit Care Med 2021;203(1):67–77; doi: 10.1164/rccm.201907-1427OC
36. De Haro C, Santos-Pulpon V, Telias I, et al.; BEARDS study investigators. Flow starvation during square-flow assisted ventilation detected by supervised deep learning techniques. Crit Care 2024;28(1):75; doi: 10.1186/s13054-024-04845-y
37. Esperanza J, Sarlabous L, de Haro C, et al. Monitoring asynchrony during invasive mechanical ventilation. Respir Care 2020;65(6):847–869; doi: 10.4187/respcare.07404
38. Tonelli R, Protti A, Spinelli E, et al. Assessing inspiratory drive and effort in critically ill patients at the bedside. Crit Care 2025;29(1):339; doi: 10.1186/s13054-025-05526-0
39. Parshall M, Schwartzstein R, Adams L, et al.; American Thoracic Society Committee on Dyspnea. An official American thoracic society statement: update on the mechanisms, assessment, and management of dyspnea. Am J Respir Crit Care Med 2012;185(4):435–452 15; doi: 10.1164/rccm.201111-2042ST
40. Manning H, Shea S, Schwartzstein R, et al. Reduced tidal volume increases 'air hunger' at fixed PC_{O_2} in ventilated quadriplegics. Respir Physiol 1992;90(1):19–30; doi: 10.1016/0034-5687(92)90131-f
41. Lansing R, Gracely R, Banzett R. The multiple dimensions of dyspnea: review and hypotheses. Respir Physiol Neurobiol 2009 May 30;167(1):53–60; doi: 10.1016/j.resp.2008.07.012
42. Grush K, Svoboda E, Dunbar P, et al. Dyspnea among mechanically ventilated patients: A systematic review. Crit Care Med 2025;53(6):e1282–e1291; doi: 10.1097/CCM.0000000000006664
43. Demoule A, Decavele M, Antonelli M, et al. Dyspnoea in acutely ill mechanically ventilated adult patients: an ERS/ESICM statement. Intensive Care Med 2024;50(2):159–180; doi: 10.1007/s00134-023-07246-x
44. Demoule A, Hajage D, Messika J, et al.; REVA Network (Research Network in Mechanical Ventilation). Prevalence, intensity, and clinical impact of dyspnea in critically ill patients receiving invasive ventilation. Am J Respir Crit Care Med 2022;205(8):917–926; doi: 10.1164/rccm.202108-1857OC
45. Schmidt M, Demoule A, Polito A, et al. Dyspnea in mechanically ventilated critically ill patients. Crit Care Med 2011;39(9):2059–2065; doi: 10.1097/CCM.0b013e31821e8779
46. Kolcak B, Ayhan H, Tastan S. The effect of using illustrated materials for communication on the anxiety and comfort of cardiac surgery patients receiving mechanical ventilator support: A randomized controlled trial. Heart Lung 2023;59:157–164; doi: 10.1016/j.hrtlng.2023.02.005
47. Telias I, Junhasavasdikul D, Rittayamai N, et al. Airway occlusion pressure as an estimate of respiratory drive and inspiratory effort during assisted ventilation. Am J Respir Crit Care Med 2020;201(9):1086–1098; doi: 10.1164/rccm.201907-1425OC
48. Schmidt M, Kindler F, Gottfried S, et al. Dyspnea and surface inspiratory electromyograms in mechanically ventilated patients. Intensive Care Med 2013;39(8):1368–1376; doi: 10.1007/s00134-013-2910-3
49. Decavale M, Bureau C, Campion S, et al. Interventions relieving dyspnea in intubated patients show responsiveness of the mechanical ventilation-respiratory distress observation scale. Am J Respir Crit Care Med 2023;208(1):39–48; doi: 10.1164/rccm.202301-0188OC
50. Persichini R, Gay F, Schmidt M, et al. Diagnostic accuracy of respiratory distress observation scales as surrogates of dyspnea self-report in intensive care unit patients. Anesthesiology 2015;123(4):830–837; doi: 10.1097/ALN.0000000000000805
51. Le Marec J, Hajage D, Decavèle M, et al. High airway occlusion pressure is associated with dyspnea and increased mortality in critically ill mechanically ventilated patients. Am J Respir Crit Care Med 2024;210(2):201–210; doi: 10.1164/rccm.202308-1358OC
52. Hashimoto H, Yoshida T, Firstiogusran A, et al. Asynchrony injures lung and diaphragm in acute respiratory distress syndrome. Crit Care Med 2023;51(11):e234–e242; doi: 10.1097/CCM.0000000000005988
53. Coiffard B, Dianti J, Telias I, et al. Dyssynchronous diaphragm contractions impair diaphragm function in mechanically ventilated patients. Crit Care 2024;28(1):107; doi: 10.1186/s13054-024-04894-3
54. Jubran A, Laghi F, Grant BJ, et al. Air hunger far exceeds dyspnea sense of effort during mechanical ventilation and a weaning trial. Am J Respir Crit Care Med 2025;211(3):323–330; doi: 10.1164/rccm.202406-1243OC
55. de Haro C, Magrans R, López-Aguilar J, et al.; Asynchronies in the Intensive Care Unit (ASYNICU) Group. Effects of sedatives and opioids on trigger and cycling asynchronies throughout mechanical ventilation: an observational study in a large dataset from critically ill patients. Crit Care 2019;23(1):245; doi: 10.1186/s13054-019-2531-5
56. Schmidt M, Banzett R, Raux M, et al. Unrecognized suffering in the ICU: addressing dyspnea in mechanically ventilated patients. Intensive Care Med 2014;40(1):1–10; doi: 10.1007/s00134-013-3117-3

Panel Discussion

Piraino: John, well done. On your slide for ineffective efforts, you gave a good example of a patient with clear severe air flow obstruction because the expiratory flow waveform was essentially flat. In those patients, adjusting PEEP may or may not help. If they're a PEEP absorber then likely you will increase the ability for them to trigger, but if they're not you just drive up plateau pressure. I'll let you comment on that, but also I think it's important to highlight that in patients with clear expiratory flow limitation, usually the response is simply to wean the pressure support because they're over-supported.

Davies: This would be a patient who's having trouble triggering due to the presence of intrinsic PEEP. The idea being that you raise the circuit pressure closer to the intrinsic PEEP level, making for easier triggering. There's a slide that I have of a patient where basically the flow came back to baseline. And so we asked this question at some of our courses, is there air trapping and without doing an expiratory hold they generally reply, no, there's not. What actually happened was that the airways totally collapsed resulting in the expiratory flow returning to baseline. I agree with you if it's not a triggering issue, there no point in increasing the PEEP unless it's for oxygenation. You make a good point about over assistance resulting in intrinsic PEEP. I think this situation often goes unnoticed. If over assistance is the cause of intrinsic PEEP then yes, reducing the support would be the prudent move.

Piraino: Yeah, in people with severe air flow obstruction, because the flow is blunted, it can be close to 0. Close enough that you don't consider this to be an issue. It's a good thing to focus on.

Davies: The other thing is you've got to know your patient. Is it a patient with COPD? If it's not a patient with COPD and they have no reason for obstruction, then I'd be much more cautious about turning the PEEP up.

Owens: John, thanks for the great talk. I like the emphasis on dyspnea and I also like how you said that when you go into the room, you can either fix the ventilator *or* you can give drugs to the patient. And I think if there are ways to fix the ventilator, that would be great. The only comment I have is right at the end there, you said we should try to use lung-protective strategies and I think everyone in the field has been successful at drilling 6 mL/kg into the residents' heads. And so a lot of times I still see patients who are dyspneic but they don't need 6 mL/kg. I actually think we've gone a little bit too far with it even in patients who don't need such small tidal volumes. Any comments on that?

Davies: Well, I will say that in our institution, we don't target 6 mL/kg, we target 4 to 8 mL/kg. So that gives us some wiggle room in terms of modulating dyspnea due to low tidal volumes. But I get what you're saying. This reminds me of the days of striving for the perfect blood gases - pH of 7.4, P_{aO_2} > 90 mm Hg, and P_{CO_2} of 40 mm Hg. We eventually realized that if we have to use injurious settings to get there, maybe that's not a good idea And I think the same thing applies here. If we need to turn up the tidal volume to relieve dyspnea, we go up to 8 mL/kg. We know that 12 mL/kg kills more than 6 mL/kg, but we don't know if 6 is better than 8.

Owens: It's sort of towards the end of mechanical ventilation that we get into this conversation with patients. I see patients where their ARDS is resolved or they don't have ARDS and patients are re-sedated because they feel dyspneic. I'm sure if you have a pH of 7.3 you're going to look bad and feel dyspneic. So again, I find myself approaching these patients who are dyspneic and I say, do they need the ventilator? First question. Second question is do they need low tidal volume ventilation? And a lot of times you can fix dyspnea by asking and answering those questions.

Davies: It begs the question, and we saw this a lot in COVID. You have somebody on low levels of pressure support say 5/5 they're pulling tidal volumes of 11 mL/kg. What's the first thing you should ask? Can they be extubated, right? And if they can't, what then? Some would argue for sedation and taking over ventilation to prevent patient self-inflected lung injury (P-SILI). Others may opt to let the patient keep going with this breathing pattern hoping P-SILI won't occur.

Beitler: Great talk, John. I really like that you highlighted that paper using high-flow during an SBT.[1] Adding to what Bob [Owens] said, asking what are we doing that makes this patient's experience of breathing abnormal shifts how we might approach the problem. Bypassing the upper airway receptors with an endotracheal tube is an obvious one. A benign intervention which is likely to be applied anyway after extubation, like putting a patient on high-flow, I think has great practical appeal. I'd be very interested in having Lluís [Blanch] detect asynchronies while patients are on or off high-flow and see if maybe that alleviates the asynchrony. Again, it is helpful to think of innovative ways to treat the abnormal experience of breathing that we are causing by nature of the therapies that we have to apply to support life.

Davies: There always seems to be controversy about how do we do spontaneous breathing trials. So if we add something that makes the patient more comfortable and makes them look better, are we actually doing them a favor? We would like to see how the patient is going to fare without excessive ventilatory support. So I'm not sure whether doing an SBT on high-flow nasal cannula is wise at this time. It may help the patient pass the SBT but put them at risk for extubation failure.

Beitler: I agree with you. To adopt that in clinical practice we would probably want to see testing with patients that have outcomes, like everything we're talking about at this conference. As Dean [Hess] has mentioned several times too, we need to be conscious about whether just because what we're seeing at the bedside might not look the way we want it to, is our intervention warranted or is the intervention potentially itself harmful?

Davies: I agree. I think that whenever we go to the bedside, you make a change for one reason like turning PEEP up for triggering. What's that going to do to the tidal volume? There's always consequences we have to keep in mind. So I agree with you totally. I mean, we have to be careful what we're doing at the bedside. You're doing something for a specific purpose. However, we must make sure we're not causing something injurious as a consequence.

Mireles-Cabodevila: It seems that a lot of the comments that have come out talk about the unintended consequences of us trying to make things pretty and following very dogmatic or control intentions. It made me think about the concept we have of permissive hypercapnia or permissive acidosis, where it improves the outcome of patients with asthma or severe obstructive lung disease. Perhaps the concept of permissive discordance, allowing a certain degree of discordance to exist, and above certain level you have to treat it, may make more sense as we gain more data from all the analysis that we do.

Blanch: You reminded me of a recent paper by Jubran et al,[2] which showed that even with adequate tidal volumes late in the ICU course, patients can still experience dyspnea. The author emphasized the importance of clinicians at the bedside asking specific questions to uncover potential causes—especially in cases where there are no obvious reasons for the dyspnea, yet the patient still reports it. It's an excellent paper, and the accompanying editorial by Banzett and colleagues is also outstanding.[3]

Davies: I agree that we should be asking the patients and communicating with them if possible. Unfortunately many times the patients are either too sick and too sedated which renders them unable to communicate or they can't communicate in a meaningful manner.

Beitler: On that point of when they cannot communicate, drawing a parallel from the discussion about assumed pain. I think it's similarly important that we don't assume dyspnea. One of the presenters yesterday spoke about the fact that there are lots of patients who look like they might have quite uncomfortable breathing, but it seems some patients adapt to that high respiratory rate that you might see in COVID, for example, and they don't experience much dyspnea. If that patient were on a ventilator, we probably would assume they were very dyspneic based on their breathing pattern. We have to be careful about over-treating symptoms that they don't have when our treatments themselves have adverse consequences.

Davies: I agree. Yesterday you were talking about COVID patients and somebody was asking them. . .

Owens: We use the expression that we have to get comfortable with the patients looking uncomfortable.

Davies: Yes, we use sedation at times to make the those caring for the patient comfortable, right?

Piraino: I just want to add to the point that when we can communicate with the patient that we don't just instantly focus on the inspiratory pressure. I'll give an example where I was called to assess a patient on non-invasive ventilation who looked horrible. My first question was, 'do you feel like you're getting enough support' and they weren't really sure. When I increased inspiratory pressure, they immediately recognized it made them feel worse. Next, I asked 'do you find it hard to breath out.' They again, seemed unsure. I turned down the EPAP and asked them if that was better, the patient's shoulders just relaxed, and they gave a thumbs up. We sometimes might forget that EPAP is expiratory resistance and turning it down can ease the work required to exhale. So, we shouldn't just focus on inspiratory pressure when trying to relieve discomfort because it may not be related to dyspnea. It's the difficulty we're creating for them exhaling against PEEP.

Davies: That's interesting. We do a lot of courses at the hospital with RNs, residents, and fellows. One part of the course is having them experience what it's like to be on the ventilator.

Piraino: Oh, EPAP is the worst feeling.

Davies: When I increase the PEEP slightly they really find it uncomfortable.

Piraino: Yeah, that's horrible.

Davies: When I get to just 10 of PEEP their eyes bulge and usually they take themselves off the ventilator.

Piraino: But I think a lot of clinicians focus on the inspiratory pressure when patients look uncomfortable and they don't realize that the expiratory pressure we're utilizing can be extremely uncomfortable. Unless they've worn it, they may not understand how uncomfortable too much EPAP can be when you don't need it.

Davies: Well, that's part of the assessment, right? Are they fighting to exhale or you're seeing a lot of expiratory movement? That's something that we should be assessing at the bedside.

Kriner: I'll just add one point to that. You know a lot of the home CPAP noninvasive devices, they have an expiratory relief feature to make them comfortable, something like C-flex, where it backs off with the pressure during expiration. They've learned that you're more comfortable if that pressure is alleviated and if I want you to be adherent and wear this device a whole lot . . .

Piraino: Plus that's the pressure when they're sleeping.

Davies: Well, why are we using PEEP? In most cases for oxygenation, right? However almost all institutions use 5 of PEEP as the default – whether it's an oxygenation issue or not.

Owens: Actually we're going to 6 to try to reduce our ventilator-associated events.[4]

Davies: Some of our therapists go to 8 for probably the same reason. But do we need to default to a PEEP of 5? The latest generation of ventilators are extremely sensitive and could probably function just as well with a PEEP of 0. And getting to Rich's [Branson] point of going to odd numbers. Using a PEEP of 7, for instance. It makes clinicians think. I think that clinicians have heard PEEP of 5 so often that information goes in one ear and out the other.

References

1. Le Stang V, Graverot M, Kimmoun A, et al. Nasal high flow to modulate dyspnea in orally intubated patients. Am J Respir Crit Care Med 2025; 211(4):577–586. doi: 10.1164/rccm.202405-1057OC
2. Jubran A, Laghi F, Grant BJB, et al. Air hunger far exceeds dyspnea sense of effort during mechanical ventilation and a weaning trail. Am J Respir Crit Care Med 2025;211(3):323–330. doi: 10.1164/rccm.202406-1243OC
3. Banzett RB, Schwartzstein RM, Brown R. Target for today: Air hunger. Am J Respir Crit Care Med 2025;211(3):300–301. doi: 10.1164/rccm.202501-0087ED
4. Ferrel E, Chapple KM, Calugaru LG, et al. Minor change in initial PEEP setting decreases rates of ventilator-associated events in mechanically ventilated trauma patients. Trauma Surg Acute Care Open 2020;5(1):e000455. doi: 10.1136/tsaco-2020-000455